

KATHMANDU UNIVERSITY
End Semester Examination [C]
July, 2017

Level : B.E./B.Sc./B. Tech.
Year : I

Course : MATH 101
Semester : I

Exam Roll No. :

Time: 30 mins.

F. M. : 20

Registration No.:

Date : JUL 07 2017

SECTION "A"
[10Q × 1 = 10 marks]

Fill in the blanks space(s) by most appropriate word(s) or symbol(s).

1. The column of a square matrix U are linearly independent if -----
2. The standard basis vector $\vec{e}_1 = (1, 0, 0)$, $\vec{e}_2 = (0, 1, 0)$ and $\vec{e}_3 = (0, 1, 0)$ in $\mathbb{R}^3$ are -----
3. The Eigen value(s) of matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is (are) -----
4. Suppose that $\sum a_n$ be a series of positive terms and $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} > 1$ then the series is -----series.
5. The function whose value at any number x is the greatest integer less than or equal to x is called the -----.
6. Function f(x) is continuous at a right end point $x = a$ of its domain if -----
7. The solution of differential equation $\frac{dy}{dx} = \frac{1}{1+x^2}$ at $y(0) = 1$ is $y =$ -----
8. The function $f(x) = |-x|$ defined on the set of real number is ----- function.
9. The function $y = f(x)$ has a jump discontinuity at $x = a$ if it satisfies -----condition.
10. If $\int_0^2 |1 - x| dx =$ -----.

SECTION "B"
[10 Q. × 1 = 10 marks]

11. The linear approximation of the function $f(x) = \sqrt{1+x}$ at $x = 0$ is -----
 $\left[1 + \frac{x}{2} \quad 1 - \frac{x}{2} \quad 2 + \frac{x}{2} \quad \frac{x}{2}\right]$
12. The function $f(x) = \frac{\log x}{x}$ is increasing in the interval -----
 $[(1, 2e) \quad (0, e) \quad (2, 2e) \quad (\frac{1}{e}, 2e)]$
13. If $y = a \sin x + b \cos x$, then $y^2 + \left(\frac{dy}{dx}\right)^2$ is a -----

[function of x
function of x and y]
function of y
constant function]

14. If function $f(x) = 2t + 1$ -----function.
[even odd; neither even nor odd; none of above]
15. The value of 'a' for which the matrix $\begin{bmatrix} a & 2 \\ 2 & 4 \end{bmatrix}$ is singular if -----
[$a = -1$; $a = 1$; $a \neq 1$; $a = 0$]
16. A function f is said to have -----value at $x = c$ if there exist an open interval (a, b) containing the point $x = c$ such that $f(x) \leq f(c), \forall x \in (a, b)$.
[absolute maximum; relative maximum; absolute minimum; relative minimum]
17. The oblique asymptote to the curve $y = \frac{x^2 - 3}{2x - 4}$ is -----
[$y = \frac{x}{2} + 1$; $y = \frac{x}{2} - 1$; $y = -\frac{x}{2} + 1$; $y = \frac{x}{2} + 2$]
18. The value of $\frac{d}{dx}(e^{\sqrt{x}})$ is -----
[$e^{\sqrt{x}}$; $2xe^{\sqrt{x}}$; $\frac{e^{\sqrt{x}}}{2\sqrt{x}}$; $e^{\sqrt{x+1}}$]
19. The critical point of the function $y = x^{-3/2}(x-2)$ is -----
[0, -2; -1, 0; -2, 2; 0, 2]
20. The derivative of the function $y = \int_0^{\sqrt{x}} \cos t dt$ is -----
[secx; cosx; $2x \cos x^2$; tanx]

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End Semester Examination [C]
July, 2017

JUL 07 2017

Level : B.E./B.Sc./B. Tech.
Year : I
Time : 2 hrs. 30 mins.

Course : MATH 101
Semester : I
F. M. : 55

SECTION "C"
[3Q.×7 = 21 marks]

1. Define differential coefficient of the function $y = f(x)$ at $x = a$. Prove that if a function $f(x)$ is differential at any point of its domain, then it is continuous. For what value of a and b will
- $$f(x) = \begin{cases} ax, & x < 2 \\ ax^2 - bx + 3, & x \geq 2 \end{cases} \text{ be differentiable for all values of } x? \quad [1+3+3]$$

OR

State second derivative test for local extrema values of the function $y = f(x)$. If $f(x) = x^{-\frac{2}{3}}(x-1)$, then investigate rise, fall and extreme value(s) of it and plot the graph based on above observation. [1.5+4+1.5]

2. Give a geometrical interpretation of the definite integral $\int_a^b f(x) dx$ of a function $f(x) \geq 0$ on the interval $[a, b]$. Explain the procedure to obtain the formula for calculating the value of definite integral $\int_a^b f(x) dx$ as a limit of sum. Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and bounded below by the x -axis and the line $y = x-2$. [1+3+3]
3. Define row rank of a matrix. Find the rank of coefficient and augmented matrix and obtain the parametrically represented solution of the system together with one specific solution: [1+1+5]

$$4x_1 + 8x_2 + 4x_3 + 4x_4 = 60$$

$$2x_1 + 5x_2 + 2x_3 + 3x_4 = 36$$

$$3x_1 + 7x_2 + 4x_3 + 3x_4 = 50$$

SECTION "D"
[6Q.× 4 = 24 marks]

4. Find $\frac{dy}{dx}$ (ANY TWO):
i) $y = \ln(\sin(\cos^{-1}(x)))$ ii) $x^m y^m = (x+y)^{m+n}$ iii) $x = e^{\log \cos 4\theta}$, $y = e^{\log \sin 4\theta}$
5. Evaluate the following integrals (ANY TWO):
i) $\int \frac{\ln x}{x+4x \ln^2 x} dx$ ii) $\int_0^{\pi/2} \frac{d\theta}{2+\cos \theta}$ iii) $\int_0^\infty \frac{\ln x}{x^2} dx$
6. What is the asymptote of the function $f(x) = c$. Determine whether the asymptote(s) to the graph of $y = \frac{x^2-4}{x^2-3}$ exist, if so find them and sketch the curve.
7. Find the volume of the solid generated by revolving the region between the parabola $x = y^2+1$ and the line $x = 3$ about the line $x = 3$.

OR

Evaluate the integral $\int_1^5 \left(\frac{x}{2} + 1\right) dx$ by using the circumscribed rectangles method.

8. Define basis. Find a basis for the subspace S of $\mathbb{R}^3$ that is the plane with equation: $6x + 5y + 2z = 0$.
9. Decide whether the series $\sum_{n=1}^{\infty} \frac{n2^n(n+1)!}{3^n n!}$ converges or diverges.

SECTION "E"

[5Q. $\times$ 2 = 10 marks]

10. When a circular plate of metal is heated in an oven, its radius increases at the rate of 0.01cm/min. At what rate is the plate's area increasing when radius is 50 cm.
11. If $f(x) = \sqrt{x-1}$, find the value of 'c' given by Mean Value Theorem on the interval $[1, 3]$.
12. Evaluate limit: $\lim_{x \rightarrow 1^+} (x-1)^{1-x}$
13. Prove that the set $S = \{(x, y, 0) : x, y \in \mathbb{R}\}$ is a vector subspace of $\mathbb{R}^3$.
14. Show that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x + y, x - y)$ is linear mapping.